

1. Prepare each file and add explicit links

VCF inputs (three separate arms)

1: Five genomes: NB72462M, NG131FQA1I, 0GOOR_HG002, HG004_GRCh38, HG005_GRCh38 (all .vcf.gz)

2: 104 single-sample cohort VCFs

3: Whole HG005_GRCh38 VCF

+ ClinVar; + frequencies (arm 2)

Then normalize (GRCh38).

VCF-RDFizer conversion

VCF → RDF graph per file.

Link plug-in: same VCF

Rules: linker.ttl

Pinned GRCh38 references

SPDI from position + alleles

Gene interval overlap

→ separate .links.nt.

RDF graphs + their link triples

Shared variant identity:

genome call → SPDI ← ClinVar call

frequency-panel call → the same SPDI

Edges: *vcfl:sameVariantAs*

Gene links for policy selection:

genome call → Ensembl gene identifier

Edge: *vcfl:overlapsGene*

Shared identifiers let queries join the files.

2. Build a checked release

Where the policy is defined

Case JSON + gene list

→ generate.py → policy.ttl

ODRL file consents and withdrawal;

28 cancer genes: clinical care only.

Purposes use DUO terms.

Policy plug-in: evaluate

Select file records and gene overlaps.

Match requester and purpose.

A prohibition overrides permission.

No permission → withhold.

Write permitted data and links.

Request context

--assignee: requester identifier

--purpose: DUO purpose term

Clinical lab / cardio study / biobank
(from the case JSON).

filtered graph

Policy plug-in: check

Compare with a VCF-derived reference.

Check ownership and dangling links.

Pass → checked view.nt.gz

One release graph per requester.

shared annotation RDF + links

3. Query the permitted graph

SPARQL query enters here

carriers.rq: 81 ACMG genes;

ALT carrier; PASS/missing FILTER;

ClinVar review; join calls by SPDI.

rare.rq: add frequency < 1%.

QLever indexes and queries

Checked participant release graph

+ ClinVar RDF and links

+ frequency RDF and links (arm 2).

Run the same query for each requester.

Returned information

File, sample, gene, chromosome,
position, REF/ALT, SPDI,
ClinVar classification, genotype;
and frequency for rare.rq.

Policy enforcement happens when the release graph is built. Requester identity and purpose are evaluation inputs; the biological query contains the analysis filters.